

Chat-GPT: CPM + Derive Test Set

Feature 1: User Course Visibility

Input Parameters

- User Role
- Enrollment Status

Categories and Choices

- **User Role:** student (valid), teacher (valid), admin (invalid), guest (invalid)
- **Enrollment Status:** enrolled (valid), not enrolled (invalid)

Test Cases

- TC1: student, enrolled → Courses listed
- TC2: teacher, enrolled → Courses listed
- TC3: student, not enrolled → Empty list
- TC4: guest, enrolled → Access denied

Feature 2: Course Access and Class Ordering

Input Parameters

- User Role
- Course Access
- Class Availability

Categories and Choices

- **User Role:** student (valid), teacher (valid), guest (invalid)
- **Course Access:** allowed (valid), denied (invalid)
- **Class Availability:** classes exist (valid), no classes (edge case)

Test Cases

- TC1: student, access allowed, classes exist → List shown
- TC2: teacher, access allowed, no classes → Empty message
- TC3: guest, access denied → Access error

Feature 3: Course Creation

Input Parameters

- User Role
- Course Title
- Course Image

Categories and Choices

- **User Role:** teacher (valid), student/guest (invalid)
- **Course Title:** provided (valid), empty (invalid)
- **Course Image:** attached (valid), not attached (valid), unsupported format (invalid)

Test Cases

- TC1: teacher, title provided, image attached → Course created
- TC2: teacher, title empty, image attached → Error on title
- TC3: student, title provided → Access denied

Feature 4: Class Creation

Input Parameters

- User Role
- Title
- Description
- DateTime
- Invited Students
- Related Files
- Forum Enabled

Categories and Choices

- **User Role:** teacher (valid), others (invalid)
- **Title:** non-empty (valid), empty (invalid)
- **Description:** provided (valid), empty (valid)
- **DateTime:** future (valid), past (invalid), malformed (invalid)
- **Invited Students:** valid emails (valid), invalid emails (invalid)
- **Files:** supported (valid), unsupported (invalid)
- **Forum:** enabled, disabled

Test Cases

- TC1: teacher, all valid inputs, forum enabled → Class created
- TC2: teacher, past date → Date error
- TC3: teacher, invalid email list → Invite error
- TC4: guest, valid data → Access denied

Feature 5: Student Interacts with Courses on Dashboard

Input Parameters

- User Role
- Forum Enabled
- Attached Files

Categories and Choices

- **User Role:** Student (valid), Teacher/Guest (invalid)
- **Forum Enabled:** True, False (valid), Null (invalid)
- **Attached Files:** Present, Absent (valid), Corrupted (invalid)

Test Cases

- TC1: Student, forum enabled, files present → View, comment, download
- TC2: Student, forum disabled, files present → View, download
- TC3: Student, forum enabled, no files → Comment only
- TC4: Teacher, forum enabled, files present → Access denied

Feature 6: Teacher Conducts Classes

Input Parameters

- User Role
- Class Started
- Audio/Webcam Control
- Chat Enabled
- Students Present

Categories and Choices

- **User Role:** Teacher (valid), Student (invalid)
- **Class Started:** Yes (valid), No (invalid)
- **Audio/Webcam:** Share, Block (valid), Unrecognized (invalid)
- **Students Present:** At least one (valid), None (edge case)

Test Cases

- TC1: Teacher, class started, share, students present → Full control
- TC2: Teacher, class not started → Error: cannot begin session
- TC3: Student, class started → Access denied
- TC4: Teacher, class started, block, no students → View only

Feature 7: Student Attends Classes

Input Parameters

- User Role
- Class Status
- Permissions Granted

Categories and Choices

- **User Role:** Student (valid), Teacher/Guest (invalid)
- **Class Status:** Started (valid), Not Started (invalid)
- **Permissions:** Granted (valid), Denied (valid), Undefined (invalid)

Test Cases

- TC1: Student, class started, permission granted → Full access
- TC2: Student, class not started → Entry blocked
- TC3: Student, class started, permission denied → View only
- TC4: Teacher, class started → Access denied

Feature 8: Users Register with Email

Input Parameters

- Email
- CAPTCHA

Categories and Choices

- **Email:** Valid (format correct), Invalid (empty/malformed)
- **CAPTCHA:** Passed (valid), Failed (invalid), Skipped (invalid)

Test Cases

- TC1: Valid email, CAPTCHA passed → Registration successful
- TC2: Invalid email, CAPTCHA passed → Error: invalid email
- TC3: Valid email, CAPTCHA failed → Error: failed CAPTCHA
- TC4: Empty email, CAPTCHA skipped → Error: fields required

Feature 9: Users Log In

Input Parameters

- Email
- Password

Categories and Choices

- **Email:** Registered (valid), Unregistered/empty (invalid)
- **Password:** Correct (valid), Incorrect/empty (invalid)

Test Cases

- TC1: Registered email, correct password → Login success
- TC2: Registered email, wrong password → Error
- TC3: Unregistered email → Email not found
- TC4: Empty credentials → Error: required fields

Feature 10: Teacher Edits Course

Input Parameters

- User Role
- Course ID
- New Title or Image

Categories and Choices

- **User Role:** Teacher (valid), Student/Guest (invalid)
- **Course ID:** Existing (valid), Non-existing (invalid)
- **Attributes:** Proper values (valid), Empty/Corrupted (invalid)

Test Cases

- TC1: Teacher, valid course ID, title updated → Success
- TC2: Teacher, invalid course ID → Error: not found
- TC3: Student, valid course ID → Access denied

Feature 11: Teacher Edits Class

Input Parameters

- User Role
- Class ID
- New Title or Image

Categories and Choices

- **User Role:** Teacher (valid), Student/Guest (invalid)
- **Class ID:** Existing (valid), Non-existing (invalid)
- **Attributes:** Proper values (valid), Empty/Corrupted (invalid)

Test Cases

- TC1: Teacher, valid class ID, title updated → Success
- TC2: Teacher, invalid class ID → Error: not found
- TC3: Student, valid class ID → Access denied

Feature 12: User Modifies Profile

Input Parameters

- User ID
- Username
- Email
- Password
- Image

Categories and Choices

- **Username:** Non-empty string (valid), Empty or symbols only (invalid)
- **Email:** Valid format (valid), Wrong format (invalid)
- **Password:** Strong password (valid), Empty or weak (invalid)
- **Image:** JPG, PNG (valid), EXE or corrupted file (invalid)

Test Cases

- TC1: Valid username, email, password, image → Profile updated
- TC2: Empty username, valid other fields → Error: Username required
- TC3: Invalid email, valid other fields → Error: Invalid email

Feature 13: Teacher Adds Course Description

Input Parameters

- User Role
- Course ID
- Description Content

Categories and Choices

- **User Role:** Teacher (valid), Student/Guest (invalid)
- **Description Content:** Text, Image, Link (valid), Executable file or invalid URL (invalid)

Test Cases

- TC1: Teacher adds text description → Description added
- TC2: Teacher adds image and link → Description added
- TC3: Teacher uploads executable file → Error: Invalid format
- TC4: Student attempts to add description → Access denied

Feature 14: CAPTCHA Required for Registration

Input Parameters

- Email
- CAPTCHA

Categories and Choices

- **Email:** Valid format (valid), Empty or malformed (invalid)
- **CAPTCHA:** Passed (valid), Failed or skipped (invalid)

Test Cases

- TC1: Valid email, CAPTCHA passed → Registration successful
- TC2: Valid email, CAPTCHA failed → Error: CAPTCHA failed
- TC3: Invalid email, CAPTCHA passed → Error: Invalid email
- TC4: Empty email, CAPTCHA skipped → Error: All fields required

Feature 15: Calendar View for Classes

Input Parameters

- User Role
- Class Data

Categories and Choices

- **User Role:** Authenticated (valid), Guest (invalid)
- **Class Data:** Exists (valid), Empty (valid but edge case)

Test Cases

- TC1: Student, class data exists → Calendar displays classes
- TC2: Teacher, no class data → Calendar empty
- TC3: Guest, class data exists → Access denied

Feature 16: Teacher Adds Students to a Course

Input Parameters

- Input Method
- Input Data Format

Categories and Choices

- **Input Method:** Single email, Multiple emails, File upload
- **Data Format:** Valid (emails properly formatted, supported file types), Invalid (malformed emails, unsupported/corrupted files)

Test Cases

- TC1: Single email, valid format → Student added
- TC2: Multiple emails, invalid format → Error: Malformed email list
- TC3: File upload (.csv), valid format → Students added
- TC4: File upload (.exe), invalid format → Error: Invalid file format